

LLMs and Corpora

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Abstract

This article examines the foundational relationship between large language models (LLMs) and text corpora. It traces how modern LLMs leverage unprecedented scales of web-crawled data—trillions of tokens from diverse sources—to acquire linguistic competence through self-supervised learning. The article covers corpus requirements, processing pipelines, pre-training objectives, and post-training adaptations, while addressing methodological considerations including reproducibility, legal implications, and emerging trends like synthetic data generation and multimodal corpora that will shape future language model development.

Keywords

Artificial intelligence, computational linguistics, corpus linguistics, large language models, machine learning, natural language processing, pre-training, self-supervised learning, text corpora, transformer models

Key Points

- Modern LLMs require corpora of unprecedented scale (trillions of tokens) from diverse sources including web crawls, books, reference content, and code
- Self-supervised learning objectives enable models to learn from unlabeled text without manual annotation
- Corpus processing involves extensive filtering, deduplication, and tokenization to prepare training data
- LLMs' pattern learning connects to foundational corpus linguistics concepts, with models internalizing distributional regularities at massive scale
- LLM outputs reflect patterns learned from training corpora but cannot be treated as direct corpus evidence. Researchers must account for training data composition, temporal cutoffs, and transparency limitations

- Emerging trends include increased use of synthetic data and multimodal corpora

Glossary

Alignment: The process of training language models to behave according to human values, preferences, and safety standards, typically through techniques like RLHF.

Corpus (plural: corpora): A large, structured collection of texts used for linguistic analysis or training computational models.

Pre-training: The initial phase where language models learn general linguistic patterns from vast unlabeled text collections.

RLHF (Reinforcement Learning from Human Feedback): A training method that uses human preference data to align model outputs with desired behaviors.

Self-supervised learning: A machine learning approach where training signals are derived from the input data itself, without manual labels.

Web crawling: The automated process of systematically browsing, downloading, and extracting content from websites for indexing or analysis.

Introduction

Large language models (LLMs) like GPT, Claude, Llama and Gemini represent a paradigm shift in how computational systems process and generate natural language. These models learn from massive text corpora through self-supervised learning, acquiring linguistic competence that enables generation of coherent, contextually appropriate text across diverse domains. Unlike traditional Natural Language Processing (NLP) systems that relied on annotated datasets and manual feature engineering, modern LLMs leverage web-scale corpora containing trillions of tokens.

The relationship between LLMs and corpora is foundational to their capabilities. Pre-training—the initial phase where models learn general language capabilities from vast unlabeled text corpora—enables these models to acquire syntactic patterns, semantic relationships, pragmatic conventions, and world knowledge. During pre-training, models process billions or trillions of tokens without task-specific supervision, learning to predict missing or upcoming words. This article examines how corpora are collected, processed, and utilized in LLM development, with particular attention to the computational and methodological considerations that distinguish this approach from traditional corpus linguistics.

Corpus Requirements for LLM Pre-training

Scale and Diversity

Modern LLMs require corpora of unprecedented scale. Early pre-trained models like BERT (Devlin et al., 2018) used 3 billion tokens from BookCorpus and English Wikipedia. GPT-3 (Brown et al., 2020) expanded to 300 billion tokens from CommonCrawl, WebText, books, and Wikipedia. The LLaMA series of models (Touvron et al., 2023) were trained on progressively large corpora from 1.4 trillion text tokens for LLaMA-1 to 15 trillion text tokens for LLaMA-3. Multimodal models LLaMA-4 and GPT-4 are trained on a combination of text, image, video making it difficult to count training tokens.

This scale serves multiple purposes. Large corpora contain extensive factual information, supporting the world knowledge that makes LLMs useful for downstream tasks. They also provide more diverse linguistic contexts, enabling models to learn low-frequency constructions and specialized terminology. The relationship between corpus size and model performance follows predictable scaling laws (Kaplan et al., 2020), with log-linear improvements as data increases. Figure 1 illustrates this trend, showing the progression of model performance on the MMLU (Massive Multitask Language Understanding) benchmark from pre-2022 through 2025, demonstrating how increasing model scale and training data volumes have led to steady improvements in capabilities.

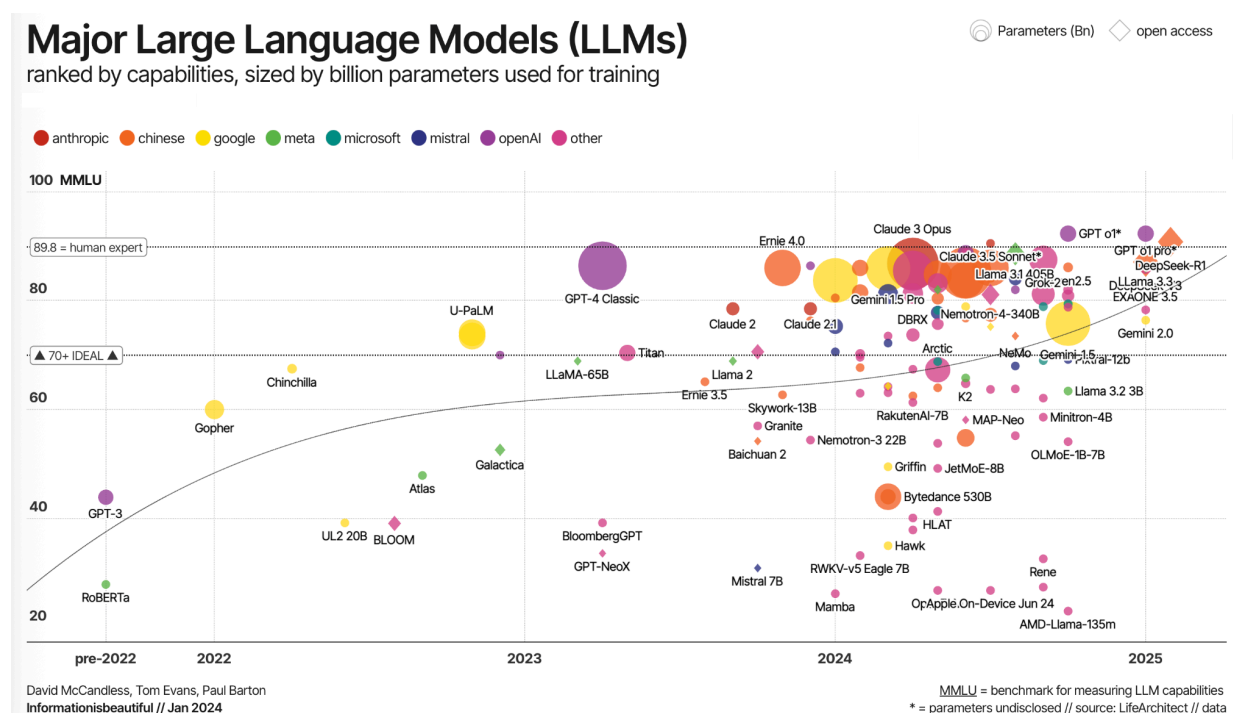


Figure 1. LLMs ranked by capabilities and sized by billion parameters used for training. The visualization shows the progression of model performance on the MMLU (Massive Multitask Language Understanding) benchmark from pre-2022 through 2025, demonstrating how increasing model scale and training data have led to improved capabilities. Source: McCandless et al. (2024).

Corpus Composition

Pre-training corpora typically combine multiple types of sources to ensure linguistic diversity and broad coverage of topics. While specific datasets vary across models, the following categories represent the main types of data used:

Web Crawls: Large-scale web data forms the backbone of most pre-training corpora.

CommonCrawl, one prominent example, provides monthly snapshots of web content totaling over 250 billion pages. The C4 corpus (Raffel et al., 2020) derives from CommonCrawl, filtered to 750GB of English text. Other models use proprietary web crawls or alternative public datasets.

Books: Books are another commonly used data source for pretraining LLMs. Examples include Project Gutenberg, an archive of 75,000 public domain books, and the Toronto BookCorpus which contains 11,000 books (985 million words) from unpublished authors.

Reference Content: Wikipedia serves as a common source of encyclopedic content, while academic repositories like ArXiv and PubMed Central contribute scientific and medical knowledge. Models may also incorporate other reference sources including news archives, educational materials, and specialized databases.

Code: Alongside natural language data, programming language data also appears in pre-training corpora. The Stack dataset, containing 430 billion tokens of code, exemplifies this category, though many models use alternative code repositories.

Corpus Processing Pipeline

Data Collection and Filtering

The foundation of modern LLM training data lies in web crawling, an automated process that systematically accesses, extracts, and stores web content. Since the late 1990s, NLP researchers have recognized the web as a valuable language data source (Resnik, 1999), establishing "The Web as Corpus" as a mainstream approach in computational linguistics (Kilgariff & Grefenstette, 2003). Organizations like Common Crawl have democratized access to web-scale data, providing free repositories of hundreds of billions of pages. This established tradition of using web-scraped data for language research laid the groundwork for today's LLM training practices.

Raw web data undergoes extensive processing before training. Initial collection via web crawling captures HTML pages, which are processed to extract textual content. Material extracted from HTML pages often includes boilerplate text, templates, error messages, pages with minimal substantive text, and other forms of repetitive or non-informative content that fails to enhance the model's language understanding or generation capabilities.

Modern corpus processing employs multiple filtering stages to identify and remove unsuitable content. Classifier-based filtering excludes web pages with text dissimilar to curated reference

sources like Wikipedia or books. Perplexity-based filtering uses language models to score text coherence and retain only the highest-scoring content. Heuristic filters identify and exclude spam, sexual content, and toxic language. Language identification ensures corpus consistency, though multilingual models retain multiple languages. State-of-the-art pipelines often combine dozens of these filtering stages to optimize training data characteristics.

Deduplication

Following content filtering, deduplication prevents models from memorizing frequently repeated content that can comprise a substantial portion of web data. Exact deduplication removes identical documents, while fuzzy deduplication identifies near-duplicates through methods like MinHash with Jaccard similarity. Lee et al. (2021) demonstrated that deduplication improves model generalization and reduces memorization of training data.

Deduplication can substantially reduce corpus size. For example, Soldaini et al. (2023) reported that the open Dolma dataset was reduced from 5.7 trillion tokens to 3 trillion tokens through aggressive deduplication. This process particularly affects templated content like privacy policies, terms of service, and boilerplate text that appears across many websites.

Tokenization

LLMs use subword tokenization schemes like Byte-Pair Encoding (Sennrich et al., 2016) or SentencePiece. These methods build vocabularies by starting with individual characters and iteratively merging the most frequent character pairs into subword units until reaching a fixed vocabulary size, typically 30,000-50,000 tokens. This approach enables models to handle rare words by decomposing them into known subwords while keeping common words intact.

Learning Objectives in Pre-training

Self-Supervised Learning Tasks

Self-supervised learning differs fundamentally from traditional supervised learning. While supervised learning requires manually annotated data (e.g., sentiment labels for movie reviews), self-supervised learning creates training signals directly from the input text. This approach enables models to learn from massive unlabeled corpora, as the "labels" are derived from the text itself. For instance, predicting the next word in a sequence requires no human annotation—the actual next word serves as the ground truth.

Pre-training employs two primary self-supervised objectives. **Causal Language Modeling** (used by GPT models) predicts the next token given preceding context, teaching models to generate coherent continuations and capture long-range dependencies. **Masked Language Modeling** (used by BERT models) predicts randomly masked tokens using bidirectional context, better capturing word relationships though not directly supporting generation.

Through these simple prediction tasks, models learn remarkably diverse capabilities. **Table 1** demonstrates that next-word prediction teaches models a variety of capabilities by allowing them to distinguish between correct and incorrect completions across various linguistic and cognitive tasks.

Table 1. Capabilities learned by language models through next-word prediction tasks. Source: Wei (2022).

Capability	Example	Correct and Incorrect(*) Predictions
Grammar	In my free time, I like to { }	{run, *banana}
Lexical semantics	I went to the zoo to see giraffes, lions, and { }	{zebras, *spoon}
World knowledge	The capital of Denmark is { }	{Copenhagen, *London}
Sentiment analysis	Movie review: I was engaged and on the edge of my seat the whole time. The movie was { }	{good, *bad}
Harder sentiment analysis	Movie review: Overall, the value I got from the two hours watching it was the sum total of the popcorn and the drink. The movie was { }	{bad, *good}
Translation	The word for "pretty" in Spanish is { }	{bonita, *hola}
Spatial reasoning	[...] Iroh went into the kitchen to make some tea. Standing next to Iroh, Zuko pondered his destiny. Zuko left the { }	{kitchen, *store}
Math question	First grade arithmetic exam: $3 + 8 + 4 = \{ \}$	{15, *11}

Pattern Learning vs. Memorization

LLMs learn statistical patterns rather than memorizing training data verbatim. During pre-training, models develop distributed representations encoding linguistic regularities across billions of parameters. Research shows models rarely reproduce training data exactly. Carlini et al. (2023) found GPT-3 could regenerate only 0.03% of training sequences when explicitly prompted. The transformer architecture's attention mechanism enables models to dynamically combine learned patterns rather than retrieve stored passages.

This pattern learning connects directly to foundational ideas in corpus linguistics. Firth's (1957) principle that "you shall know a word by the company it keeps" anticipated the distributional

approach that underlies modern LLMs. Where corpus linguists use concordances and collocation statistics to study how words pattern together in context, LLMs internalize these distributional regularities at massive scale during pre-training. The result is that LLMs develop sensitivities to collocational patterns, register variation, phraseological conventions, and genre-specific language—the same phenomena that corpus linguists have long studied through frequency analysis and concordancing.

Post-training Adaptation and Applications

From Pre-trained Models to Practical Tools

Pre-training establishes general language understanding, but post-training adaptation is what makes LLMs useful for specific tasks. Through supervised fine-tuning on curated instruction-response pairs, reinforcement learning from human feedback (RLHF), and parameter-efficient methods like LoRA (Hu et al., 2022), pre-trained models are transformed into systems that can follow instructions, engage in dialogue, and perform targeted tasks. Safety alignment procedures further shape model behavior to be helpful while avoiding harmful outputs (Bai et al., 2022).

These adaptations mean that a single pre-trained model can be configured for tasks where corpora have traditionally played a central role. LLMs can perform text classification, named entity recognition, sentiment analysis, and summarization. These tasks previously required purpose-built annotated corpora and task-specific models. They can generate translations across dozens of language pairs, identify register and genre distinctions, and annotate linguistic features at scale. For corpus linguists, this raises both opportunities and methodological questions.

Implications for Corpus-based Research

LLMs offer corpus linguists powerful new tools for tasks that have traditionally been labor-intensive. Automatic annotation of large corpora can be performed with high accuracy, reducing the manual effort required to create richly annotated corpora. LLMs can also assist in corpus compilation itself, helping to classify texts by genre, register, or domain during the collection process.

At the same time, LLMs complicate the traditional corpus linguistics methodology. Because LLMs generate fluent text that reflects aggregate patterns from their training data, their outputs can be studied as a kind of corpus-derived linguistic evidence, but one that is filtered through the model's learned representations rather than directly attesting natural language use. Researchers must consider whether patterns observed in LLM outputs reflect genuine linguistic regularities or artifacts of training data composition and model architecture. The relationship between LLMs and corpora is thus bidirectional: corpora shape LLMs, and LLMs are increasingly used to build, annotate, and analyze corpora.

Treating LLMs as Corpus-derived Resources

Because LLMs internalize patterns from their training corpora, their outputs can be treated as a form of corpus-informed linguistic evidence. Researchers using LLMs in corpus-type analyses must consider several factors that shape what knowledge these models encode and how reliably their outputs reflect natural language use.

Domain Coverage and Diversity

Corpus composition directly affects model capabilities across domains. Models trained primarily on web text like Reddit excel at informal communication but may struggle with specialized technical writing. Including academic papers improves performance on scientific tasks, while code improves logical reasoning capabilities. The Pile (Gao et al., 2020) demonstrates targeted corpus design, combining 22 datasets totaling 825GB. Components include PubMed Central (90GB), ArXiv (56GB), Stack Exchange (32GB), and USPTO patents (23GB). Models trained on The Pile show improved performance on domain-specific benchmarks compared to generic web-trained models.

More broadly, diversity of pre-training data is essential for robust model performance across tasks. Longpre et al. (2024) found that removal of large, diverse data sources like Common Crawl can have adverse effects on model performance across downstream tasks. This diversity encompasses multiple dimensions: domain diversity across scientific, literary, conversational, and technical texts; register diversity capturing formal academic writing alongside informal social media posts; temporal diversity including both historical and contemporary content; geographic and cultural diversity representing different English variants; and language coverage determining multilingual capabilities. Models trained on diverse languages develop cross-lingual transfer abilities, with corpora like the Multilingual Colossal Clean Crawled Corpus covering 101 languages (Xue et al., 2021).

For corpus linguists, this means that any analysis of LLM outputs must account for the composition of the training data. Models trained on narrow corpora exhibit degraded performance outside their training distribution, and patterns observed in LLM-generated text will inevitably reflect the domains and registers that predominate in the training

Temporal Limitations

Training corpora inherently contain information only up to their collection date, creating knowledge cutoffs. Pretrained LLMs cannot reference events, publications, or cultural developments occurring after their training data was collected. This limitation becomes more pronounced over time as the gap between training and deployment widens.

The computational cost of pre-training, often millions of dollars for large models, makes frequent retraining impractical. Continued pre-training on recent data offers a middle ground, updating models with new information without full retraining. However, this approach risks catastrophic forgetting of previously learned capabilities. For researchers treating LLMs as linguistic

resources, temporal cutoffs mean that LLM outputs may not reflect recent language change, emerging terminology, or contemporary usage patterns.

Representativeness and Size Estimation

A foundational concern in corpus linguistics is representativeness, that is, whether a corpus adequately reflects the language variety it claims to represent. This question becomes especially challenging with LLMs, where the training corpus is mediated through model parameters rather than directly accessible. Quantifying and comparing the size of LLM training corpora is itself difficult: token counts depend on tokenization schemes, deduplication and filtering dramatically reduce raw data volume, and published statistics often report post-processing sizes without specifying filtering criteria. Research on compute-optimal training (Hoffmann et al., 2022) suggests approximately 20 tokens per parameter, but such estimates offer limited insight into what linguistic phenomena are actually represented in the data.

Unlike a traditional corpus, where researchers can inspect concordance lines and verify coverage, an LLM's training data is irreversibly transformed into model weights. Researchers cannot straightforwardly determine whether a particular genre, register, or variety is well-represented in the model's knowledge. This opacity makes it difficult to assess whether patterns in LLM outputs reflect broad linguistic regularities or the idiosyncrasies of an unbalanced training set.

Reproducibility, Transparency and Legal Considerations

Many commercial LLMs lack detailed corpus documentation, which makes those models less suitable for avenues of research that would benefit from not only access to the model but also knowledge of the details of its training corpus. When the composition of the training data is unknown, the provenance of any linguistic pattern observed in model outputs is uncertain. Some organizations have released detailed corpus documentation alongside their models. The Allen Institute for Artificial Intelligence, for example, released not only the weights of its Open Language Model but also the complete Dolma training dataset with full documentation of data sources, processing code, and filtering methods. Researchers have also developed Model Cards and Data Cards that document training data characteristics, model capabilities, and intended use cases (Mitchell et al., 2019). For corpus linguists, models with documented training data offer more reliable foundations for linguistic analysis.

LLMs and the manner in which they are trained have raised myriad questions both in the courts and beyond about copyright and privacy. LLM corpora may include information protected by copyright laws, which raises the question whether the doctrine of fair use applies to the use of such content to train machine learning models. Beyond copyright, corpora may contain personal information despite filtering attempts, and corpus biases propagate to model outputs, potentially amplifying societal prejudices. These concerns are familiar to corpus linguists who have long grappled with ethical data collection, but the scale and opacity of LLM training data intensify them considerably.

Current Trends in Corpus Development

Data Exhaustion

The rapid scaling of LLMs has led to concerns about exhausting text data. Villalobos et al. (2022) project that publicly accessible text data could be exhausted between 2026 and 2032 if current trends continue. This has motivated exploration of alternative sources including transcribed audio from podcasts, parliamentary proceedings, and subtitles, though these require more processing and exhibit different linguistic properties than written text.

Synthetic Data

LLMs increasingly generate their own training data. Self-instruct methods (Wang et al., 2022) use models to create instruction-following examples. While risks include model collapse, a gradual degenerative process where models forget the natural data distribution (Shumailov et al., 2023), careful approaches can mitigate this. Teacher-student frameworks enable large LLMs to train smaller, more efficient models through model distillation (Hinton et al., 2015), with models like Alpaca (Taori et al., 2023) demonstrating that 175B parameter teachers can produce capable 7B parameter students.

Recent work shows synthetic data generation can enhance rather than degrade performance when properly managed. Alemohammad et al. (2023) demonstrate that mixing synthetic and real data in appropriate proportions avoids model collapse. Li et al. (2024) show that using multiple models to generate diverse synthetic data with appropriate filtering produces effective training sets.

Multimodal Corpora

Next-generation models integrate text with images, audio, and video. These require aligned multimodal corpora like LAION-5B (5 billion image-text pairs) or video datasets with captions. Multimodal training may enhance language understanding through grounded representations. Grounded representations refer to language representations that are connected to perceptual experiences or sensory information, helping models better understand concepts by linking them to visual, auditory, or other non-textual modalities.

The impact of scale on multimodal capabilities mirrors patterns in text-only models. **Figure 2** demonstrates how increasing model size from 350M to 20B parameters dramatically improves image generation outputs. At 350M parameters, generated images of "a kangaroo wearing an orange hoodie and blue sunglasses standing in front of the Sydney Opera House" show basic object recognition but poor composition and detail. At 20B parameters, the same prompt produces photorealistic images with accurate spatial relationships, textures, and lighting—illustrating how scale enables models to learn complex visual-linguistic correspondences.



350M parameters

750M parameters

3B parameters

20B parameters

Figure 2. Generated images show progressive output improvements with increasing model scale. The same prompt produces increasingly photorealistic and compositionally accurate images as model parameters increase. Source: Yu et al. (2022) and <https://sites.research.google/parti/>

Conclusion

LLMs are dependent on massive text corpora. While building on a history of corpus-based NLP, the precipitous increase in the scale and diversity of LLM training data has created dramatically better, general-purpose AI systems. These models learn generalizable patterns supporting flexible language use rather than retrieving memorized examples. Understanding LLM-corpus relationships remains crucial as these models become ubiquitous. Corpus decisions—what to include, how to filter, how much to collect—fundamentally shape model capabilities and limitations. As the field evolves, corpus design will remain central to advancing language model capabilities.

Declaration of Generative AI and AI-assisted technologies in the writing process

The author used Anthropic’s Claude Opus 4 to improve the readability and language of this manuscript, including editing text, formatting references, creating figure captions, suggesting relevant citations, and revising sections for clarity. The author reviewed and edited all content as needed and takes full responsibility for the content of the publication.

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Relevant Websites

Allen Institute for AI Dolma Dataset: <https://allenai.org/dolma>

Common Crawl: <https://commoncrawl.org>

Hugging Face Datasets: <https://huggingface.co/datasets>

LAION Open Datasets: <https://laion.ai/>

Papers With Code - Language Modelling: <https://paperswithcode.com/task/language-modelling>

The Pile Dataset: <https://pile.eleuther.ai>